Irin Hoque Orchi

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Portfolio | Google Scholar | GitHub | LinkedIn

Research Interests

Natural Language Processing (NLP), Machine Learning, Large Language Models (LLMs), Low-Resource and Code-Mixed NLP, Uncertainty-Aware Learning, Human-Centered AI, Annotation Disagreement, Human–AI Interaction.

Currently investigating: How displaying LLM-generated confidence scores alongside suggested labels influences human annotation decisions, particularly for ambiguous and high-disagreement items. The study examines whether confidence information affects annotators' agreement with model suggestions and their judgments of uncertain cases.

Education

BRAC University Dhaka, Bangladesh

B.Sc. in Computer Science and Engineering, GPA 3.58/4.00, Distinction *Summer 2023*

- **Honors:** Vice Chancellor's List (Spring 2020); Dean's List (Summer 2020, Spring 2021, Fall 2022); merit-based scholarship.
- **Undergraduate thesis (2022–2023):** *Mental Health Analysis of Cancer-Diagnosed Patients with the Lowest Survival Rate and Their Caregivers*. Supervisor: Dr. Farig Yousuf Sadeque.
  - Dataset of 10,087 forum posts (Reddit, DailyStrength, HealthBoard) on five cancers with survival below 60%; 2,373 posts hand-scored for four-point valence by five annotators (pairwise Cohen's κ 0.21–0.59) and adjudicated by a third scorer.
  - Four-class valence classification: GRU over custom FastText embeddings, accuracy 0.75 (macro-F1 0.71); few-shot SBERT (all-MiniLM-L6-v2), accuracy 0.71 (macro-F1 0.67).

Publications

Peer-reviewed

[C1] **I. H. Orchi**, T. Manzoor, M. W. R. Araf, M. S. Omi, T. A. Abir, F. Sadeque. "A Qualitative Study of the Mental Health Experiences of Cancer Patients with the Lowest Survival Rate and Their Caregivers." *IEEE ICCIT 2023*. DOI: 10.1109/ICCIT60459.2023.10441282

[C2] T. Manzoor, M. W. R. Araf, M. S. Omi, T. A. Abir, A. D. Abir, **I. H. Orchi**, F. B. Ashraf. "Mitigating Online Harassment: Machine Learning Approaches for Hate Speech Detection in Transliterated Bengali Comments." *IEEE ICCIT 2023*. DOI: 10.1109/ICCIT60459.2023.10441244

Under review (IEEE ICCIT 2026)

[S1] **I. H. Orchi**, T. Manzoor, T. A. Medha, A. Zaman, S. R. Ramisa. "Class Imbalance in Regional Bangla Dialect Identification: A Benchmark Study with Interpretability Analysis."

[S2] **I. H. Orchi**, T. Manzoor, A. Zaman, S. R. Ramisa, F. Sadiq. "Adaptive Clarification: Evaluating LLM Uncertainty Signals for Ambiguity-Aware Dialogue."

[S3] T. Manzoor, **I. H. Orchi**, F. Sadiq. "AegisRL: An Adaptive Reinforcement Learning Framework for Alert Fatigue Mitigation in Security Operations Centers."

Research Experience

Class Imbalance in Bangla Dialect Identification First author, under review [S1]

BanglaDial: 61,318 sentences, 12 classes

- Tested class weighting, oversampling and SMOTE with LinearSVC and logistic regression under 5-fold CV, using Nadeau–Bengio corrected resampled t-tests with Benjamini–Hochberg FDR control. None gave a significant minority-class gain; all three lowered LinearSVC macro-F1 from 0.756.
- Found 2.33% of sentences duplicated under different dialect labels, concentrated in the five-dialect cluster that dominates the confusion matrix.
- A two-stage hierarchical classifier scored 0.736 macro-F1 with learned routing and 0.797 with oracle routing, locating the bottleneck in routing accuracy.

Adaptive Clarification: LLM Uncertainty Signals for Dialogue First author, under review [S2]

AmbigNQ; Qwen2.5-1.5B/7B-Instruct

- Combined semantic entropy (DeBERTa-v3 entailment clustering) with a 17-feature ambiguity classifier into an ANSWER/HEDGE/CLARIFY policy.

- Entropy separated ambiguous questions weakly at n=60 (AUC 0.596) and near chance at n=1,200 (AUC 0.517); model and sample size changed together, so the drop is not attributed to either.
- Classifier LOO macro-F1 0.571 at n=1,200; 65.8% of CLARIFY decisions were ambiguous questions against a 48.3% base rate. Offline evaluation only.

Pandulipi: Hate Speech Detection in Transliterated Bengali Co-author, 2022–2023

Publication [C2]

- Contributed to data collection, annotation and classifier experiments for a corpus of about 5,000 Banglish comments (2,837 hate, 2,163 non-hate; six hate categories).
- Eight classifiers under TF-IDF and Bag-of-Words. Best binary: Multinomial NB with Bag-of-Words (accuracy 0.760). Best multiclass: logistic regression with Bag-of-Words (0.615).

Datasets

- **Mental Health Insights of Vulnerable Cancer Patients**: 10,087 posts, 2,373 hand-scored for valence (Kaggle, Mendeley).

Technical Skills

ML/NLP libraries: PyTorch, Keras, Hugging Face Transformers, scikit-learn, imbalanced-learn

Models and representations: TF-IDF (word and character n-grams), Bag-of-Words, FastText, SBERT, DeBERTa-v3 (NLI), Qwen2.5-Instruct (inference), GRU/BiLSTM, SVM, logistic regression

NLP and LLM methods: text classification, dialect identification, hate speech and valence classification, semantic entropy, ambiguity classification, selective prediction (ANSWER/HEDGE/CLARIFY), corpus auditing

Evaluation and annotation: Cohen's κ and adjudication, corrected resampled t-tests, Benjamini–Hochberg FDR, Cohen's d, stratified and leave-one-out CV, ROC-AUC

Programming and engineering: Python, C++, TypeScript, SQL, Git, LaTeX; ASP.NET Core, Angular, REST APIs, Firebase

Industry Experience

Software Engineer August 2024 – Present

Bdjobs.com Limited

- Develop production web applications with ASP.NET Core, Angular and RESTful APIs, contributing features from design through deployment.
- Building an AI-based candidate–job matching score that ranks candidate profiles against job requirements.

IT Intern November 2023 – February 2024

Dhaka Bank PLC

- Implemented biometric and face-recognition technology for employee attendance systems.

Presentations & Outreach

- Presented paper [C1] as first author at IEEE ICCIT 2023, Cox's Bazar, Bangladesh (December 2023).
- Delivered a talk on the Digital Equity Fellowship 2023 at Save the Children Bangladesh, addressing digital equity, ICT skills, and gender inclusion in technology.

Leadership & Activities

- Assistant Secretary, BRAC University Cultural Club (BUCUC) — 2020–2022
- Member, Photography Dept., BRAC University Adventure Club (BUAC) — 2020–2022
- Junior Executive, Finance & Marketing, Robotics Club of BRAC University (ROBU) — 2019–2022
- Volunteer, Kishor Alo, Prothom Alo — 2014–2018